Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/28/2022
PI Name:	Allen, Jeffrey S.		
Project Title:	Testing the Equivalence of Evaporation and Condensation Coefficients Using the Constrained Vapor Bubble (CVB) Data from ISS Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jstallen@mtu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	906-487-2349
Organization Name:	Michigan Technological University		
PI Address 1:	815 R.L. Smith Building		
PI Address 2:	1400 Townsend Drive		
PI Web Page:			
City:	Houghton	**State:**	MI
Zip Code:	49931	**Congressional District:**	1
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	10/30/2018	**End Date:**	10/29/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/29/2021 per NSSC information (Ed., 4/22/21)		
Key Personnel Changes/Previous PI:	NOTE (Ed., Nov 2018): This work is result of proposal by graduate student Kishan Bellur, Ph.D. candidate at Michigan Technological University, who will be performing the work with advisor PI Dr. Jeffrey Allen. August 2020 report: Wonyoung Choi (studying at Case Western Reserve University) worked on this project as a visiting undergraduate researcher during the summer of 2019. Tasks included image analysis and preliminary ANSYS modeling and UDF development. August 2019 report: Since the original proposal, Kishan Bellur has completed his PhD and is now working on this grant as a post-doc. January 2022 report: Kishan Bellur has accepted a tenure-track faculty position in the Department of Mechanical and Materials Engineering at the University of Cincinnati.		
COI Name (Institution):	Bellur, Kishan Ph.D. (Michigan Technological University)		
Grant/Contract No.:	80NSSC19K0160		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Phase change is ubiquitous and kinetic theory is widely used to describe liquid-vapor phase change, but accurate values of evaporation coefficient and condensation coefficients are still largely unknown. For water alone, decades of research has yielded phase change coefficients that span several orders of magnitude. Evaporation and condensation coefficients are essential in analyzing liquid-vapor phase change processes. Several researchers make two primary assumptions to simplify the analysis: (1) the evaporation coefficient is equal to the condensation coefficient and (2) the liquid-vapor interface temperature is equal to the vapor temperature. Both assumptions have been shown to introduce errors. A new methodology to determine the evaporation and condensation coefficients is proposed using the Constrained Vapor Bubble (CVB) experimental data. Interferometric image analysis of the CVB data enables the reconstruction of the liquid-vapor interface to account for enhanced evaporation at the contact line. The thermal data from the experiments allows for the determination of the outer wall heat transfer characteristics and the net heat transfer to the vapor bubble. The fact that the CVB cell is a closed system allows for additional conservation equations such that the common assumptions stated above are not made. The CVB configuration along with the interferometric images, pressure, and thermal data allows for the investigation of common assumptions made in kinetic theory and for the first time, determine both the evaporation and condensation coefficients from a coupled analysis. The methodology proposed is not limited to the fluid and is applicable to any pure wetting/partially wetting fluid in the CVB setup. In addition to the development of a new methodology, the proposed work has the potential to address fundamental questions regarding commonly made assumptions in the kinetic theory of phase change. The results of the proposed research will have far reaching implications beyond space technology to applications including aerosol transport in the atmosphere, atmospheric science, and climate, micro- and nano-scale thermal transport in microelectromechanical systems (MEMS) applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Evaporation and condensation is ubiquitous, but a fundamental understanding is still lacking. This is in part due to the lack of knowledge of phase change coefficients commonly used in kinetic theory of gases. For closure and simplicity, several simplifying assumptions are generally made even though they are known to be suspect. The current work utilizes high quality data from International Space Station (ISS) experiments to test these simplifying assumptions. In addition, a new model of liquid-vapor phase change has been developed using commercial computational fluid dynamics (CFD) software in CFD. A novel “active surface” technique has been added to this software. The results of the proposed research will have impact beyond space technology to applications such as aerosol transport in the atmosphere, atmospheric science, and climate science, as well as micro and nano-scale thermal transport in microelectromechanical systems (MEMS) applications.
Task Progress:	The Constrained Vapor Bubble (CVB) experiment comprises a self-contained system that allows for closure of conservation equations so that typical simplifying assumptions are not necessary. The CVB configuration along with the interferometric images, pressure and thermal data allows for the investigation of common assumptions made in kinetic theory and for the first time, determines both the evaporation and condensation coefficients from a coupled analysis. Because the experiments are closed and without the influence of gravity, the CVB data provides a unique opportunity to isolate and investigate the various implementations of kinetic theory for predicting rates of phase change. The research effort is divided into two phases. The first focuses on analyzing CVB data. Interferometric image analysis enables the reconstruction of the liquid-vapor interface and was the central topic of the 2019-2020 annual report. Since then, advances have been made to significantly increase the signal-to-noise ratio and extrapolate the profiles to span multiple length scales (nanometer to millimeter). Reconstruction of the entire 3D liquid-vapor surface was incomplete since the image data from the cooler region of the CVB was missing from the Physical Sciences Informatics (PSI) database. Further, thermal data needed to determine the net heat input to the vapor and the outer wall heat transfer characteristics were also missing from the PSI database. Using data obtained directly from Prof. Plawsky’s research group, the thermal analysis has been completed. The second phase encompasses the primary objective of the project and is the focus of the current annual report. Building on the data generated during the first phase, a multiscale model of phase change using kinetic theory has been developed and published [Bellur et al., 2020] and a demonstration of the ability to determine accommodation coefficients from such an approach is being prepared for submission. A mass balance in the vapor bubble is used to determine the ratio of evaporation to condensation coefficients, and an energy balance is used to determine the condensation coefficient. Interfacial temperatures are solved iteratively using a custom CFD model built in ANSYS FLUENT. The local mass flux across the liquid-vapor interface is implemented as a non-uniform mass source/sink with a no-slip boundary condition at all liquid-solid interfaces. Thermal contribution of phase change is modeled as heat source/sink at all points on the liquid-vapor interface with the experimental heater power, cooler temperature, and the outer wall heat transfer coefficient as thermal boundary conditions. Iteration of interfacial temperatures is continued until convergence is achieved in both the condensation coefficient and the ratio of evaporation to condensation coefficients. The methodology is established and tested with a complete 3D interface geometry extracted from phase 1. However, we have discovered that a complex wavy Marangoni flow pattern exists even at low heat fluxes. These flow patterns corroborate with experimental interferometric images. The flow patterns change the dynamics of phase change and local interface shape. The result is an instability in the iterative process to determine the coefficients. The original goal was to develop a model that could extract both evaporation and condensation coefficients from the CVB ISS data. To that end, the modeling methodology is feasible and demonstrates the capability to resolve both coefficients. In order to accurately compute the values, however, a transient/flexible interface approach may be necessary, which is beyond the scope of the current project. Higher resolution experimental data, both spatially and temporally, is also needed. Reference K. Bellur, E. F. Médici, C. K. Choi, J. Hermanson, and J. S. Allen. “A multi-scale approach to model steady meniscus evaporation”, Physical Review Fluids, 5(2), 024001. 2020.
Bibliography Type:	Description: (Last Updated: 05/04/2026)

Abstracts for Journals and Proceedings	Bellur K, Allen JS. "Testing fundamental assumptions using Constained Vapor Bubble (CVB) data from ISS experiments." Oral presentation. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5 – 6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5 – 6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Bellur K, Médici EF, Hermanson J, Choi CK, Allen JS. "Guess free evaporation modeling without adsorbed film constraints." Surface Wettability Effects on Phase Change Workshop (Oral/Poster Presentation), Brighton U.K., October 6 – 8, 2021. Surface Wettability Effects on Phase Change Workshop, Brighton U.K., October 6 – 8, 2021. , Oct-2021
Articles in Peer-reviewed Journals	Bellura K, Medici EF, Hussey DS, Jacobson DL, LaManna J, Leao JB, Scherschligt J, Hermanson JC, Choi CK, Allen JS. "Data from cryo-neutron phase change experiments with LH2 and LCH4." Data Br. 2022 Aug;43:108474. https://doi.org/10.1016/j.dib.2022.108474 , Aug-2022
Articles in Peer-reviewed Journals	Bellura K, Medici EF, Hussey DS, Jacobson DL, LaManna J, Leao JB, Scherschligt J, Hermanson JC, Choi CK, Allen JS. "Results from neutron imaging phase change experiments with LH2 and LCH4." Cryogenics (Guildf). 2022 Jul;125:103517. https://doi.org/10.1016/j.cryogenics.2022.103517 , Jul-2022
Articles in Peer-reviewed Journals	Chakrabarti U, Yasin A, Bellur K, Allen JS. "An investigation of phase change induced Marangoni-dominated flow patterns using the Constrained Vapor Bubble data from ISS experiments." Frontiers in Space Technologies. 2023 Nov 14;4:1263496. https://doi.org/10.3389/frspt.2023.1263496 , Nov-2023
Articles in Peer-reviewed Journals	Bellur K, Medici EF, Hermanson J, Choi CK, Allen JS. "Modeling liquid–vapor phase change experiments: Cryogenic hydrogen and methane." Colloids and Surfaces A: Physicochemical and Engineering Aspects. 2023 Oct 20;675:131932. https://doi.org/10.1016/j.colsurfa.2023.131932 , Oct-2023